

# LIFTING OPERATIONS

Control dropped-load, crushing and equipment-failure hazards.

## PURPOSE AND SCOPE

Review dropped-load, crushing, impact, overturning, overloading and communication hazards during lifting operations, and verify the lifting plan, competent personnel, equipment suitability and work-area controls before starting. A lifting operation involves more than the crane itself. The operator, rigger, signalman, lifting accessories, ground conditions, wind, load centre of gravity and work area must be considered together.

## KEY HAZARDS

|                                                                                                                  |                                                                                                                   |                                                                                                             |                                                                                                              |
|------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <p>1</p>  <p>SUSPENDED LOAD</p> | <p>2</p>  <p>DROPPED LOAD</p>    | <p>3</p>  <p>CRUSHING</p> | <p>4</p>  <p>CAPACITY</p> |
| <p>5</p>  <p>RIGGING</p>      | <p>6</p>  <p>COMMUNICATION</p> | <p>7</p>  <p>GROUND</p> | <p>8</p>  <p>WIND</p>   |

## CRITICAL VALUES TO VERIFY IN THE FIELD

|                                                                 |                                                                               |                                                      |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------|
| <p><b>100%</b></p> <p>Load weight and capacity verification</p> | <p><b>1 signal person</b></p> <p>Designated person directing the operator</p> | <p><b>0</b></p> <p>People below a suspended load</p> |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------|------------------------------------------------------|



### REMEMBER - SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

# LIFTING OPERATIONS

Control dropped-load, crushing and equipment-failure hazards.

## STANDARDS AND FIELD REQUIREMENTS

OSHA 1926 Subpart CC and 1926.251 address lift planning, capacity, rigging, signal persons, exclusion zones and operational control.

### OSHA 1926.1417

#### SAFE OPERATION

Follow manufacturer procedures, load chart, level, configuration and operating limitations.

### OSHA 1926.1425

#### SUSPENDED LOAD

Keep employees clear of the fall zone and allow controlled access only for essential personnel.

### OSHA 1926.1428

#### SIGNAL-PERSON QUALIFICATION

Use a qualified signal person where required and verify the communication method before lifting.

### OSHA 1926.251

#### RIGGING INSPECTION

Inspect slings and hardware each shift and before use; remove defective gear from service.

## MINIMUM EVIDENCE BEFORE START

- |                                                                 |                                                            |
|-----------------------------------------------------------------|------------------------------------------------------------|
| <input type="checkbox"/> Approved lifting plan                  | <input type="checkbox"/> Load weight and center of gravity |
| <input type="checkbox"/> Crane and rigging certificates         | <input type="checkbox"/> Pre-shift inspection              |
| <input type="checkbox"/> Signal-person and rigger qualification | <input type="checkbox"/> Exclusion-zone plan               |

## SUPERVISOR DISCUSSION POINTS

- Has the load weight and centre of gravity been confirmed?
- Is the selected equipment capacity suitable?
- Who are the operator, rigger and signalman?

## TRIAL LIFT AND LOAD CONTROL



Plan - Capacity - Rigging - Trial lift - Tag line - Exclusion zone

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## CRITICAL CONTROL MEASURES (HIERARCHY OF CONTROLS)

### 1. PLAN THE LIFT

Verify the approved lifting plan and risk assessment. Confirm the load weight and centre of gravity.

### 2. VERIFY EQUIPMENT AND GROUND

Check crane and accessory capacities. Inspect slings, hooks, shackles and other equipment. Assign competent operators, riggers and signalmen.

### 3. CONTROL THE LOAD

Barricade the lifting area and exclude personnel. Provide reliable communication for blind lifts. Use tag lines where appropriate.

### 4. MANAGE AREA AND COMMUNICATION

Complete a trial lift to confirm load stability. Verify wind, ground and environmental conditions.

### CRITICAL WARNING



**No one may stand below a suspended load or in the line of fire. Stop the lift whenever there is doubt.**

## SITE CHECKLIST (BEFORE STARTING WORK)

- |                                                       |                                                      |
|-------------------------------------------------------|------------------------------------------------------|
| <input type="checkbox"/> Lifting plan approved        | <input type="checkbox"/> Load weight confirmed       |
| <input type="checkbox"/> Centre of gravity identified | <input type="checkbox"/> Equipment capacity suitable |
| <input type="checkbox"/> Accessories inspected        | <input type="checkbox"/> Personnel competent         |
| <input type="checkbox"/> Area barricaded              | <input type="checkbox"/> Communication ready         |
| <input type="checkbox"/> Trial lift planned           |                                                      |

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## ATTENDANCE AND APPROVAL

Project / Site: \_\_\_\_\_ Date: \_\_\_\_\_

Presented by: \_\_\_\_\_ Work area: \_\_\_\_\_

| No | Full Name | Company / Role | Signature |
|----|-----------|----------------|-----------|
| 1  |           |                |           |
| 2  |           |                |           |
| 3  |           |                |           |
| 4  |           |                |           |
| 5  |           |                |           |
| 6  |           |                |           |
| 7  |           |                |           |
| 8  |           |                |           |
| 9  |           |                |           |
| 10 |           |                |           |
| 11 |           |                |           |
| 12 |           |                |           |